

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

8 Data Science (djw1005)

- (a) In a COVID vaccine trial, n_0 subjects were given a placebo and n_1 were given the vaccine; x_0 of the placebo subjects developed the disease and x_1 of the vaccinated subjects. Considering the probability model $X_k \sim \text{Binomial}(n_k, p_k)$, the vaccine efficacy is defined to be $e = 1 - p_1/p_0$.
- (i) State the maximum likelihood estimators for p_0 and p_1 . Give a formula for the maximum likelihood estimator for e . [2 marks]
- (ii) Explain how to compute a 95% confidence interval for e . Also explain how to test whether $e > 0.5$. [7 marks]
- (b) Further data about the trial has been made available, and we learn that subjects weren't all enrolled for the same length of time. We are given a full dataset consisting of three features, the predictor variable d_i and the response variables (t_i, c_i) for subject i . Here $d_i = 1$ if the subject received the vaccine and $d_i = 0$ otherwise; $c_i = 1$ if the subject developed the disease and $c_i = 0$ otherwise; and t_i is the day on which the subject developed the disease if $c_i = 1$, and the number of days enrolled in the trial otherwise.

Consider the following probability model. Among vaccinated subjects, the vaccine is effective with probability f and ineffective otherwise. Effectively vaccinated subjects never get the disease. For ineffectively vaccinated subjects, and for subjects on placebo, each day there is a probability q of developing the disease. The parameters f and q are unknown.

- (i) For a subject i who received placebo, give an expression for the likelihood of the pair (t_i, c_i) . [4 marks]
- (ii) For a subject i who received the vaccine, give an expression for the likelihood of the pair (t_i, c_i) . [4 marks]
- (iii) Give an expression for the log likelihood of the entire dataset. [3 marks]

[Note: Your answers to this question should be symbolic. But you may like to know that in the low-dose part of the Oxford–AstraZeneca trial, $n_0 = 1374$, $n_1 = 1367$, $x_0 = 30$, and $x_1 = 3$.]

Answer:

— *Solution notes* —

Part (a)(i). $\hat{p}_k = x_k/n_k$ and $\hat{e} = 1 - \hat{p}_1/\hat{p}_2$. Students should be able to just write down the answer.

Part (a)(ii). This is just like every other confidence interval and hypothesis test question. For the hypothesis test, the answer needs to give a reasonable test statistic, and a reasonable choice of sidedness. It's important to get the sampling right: sample under the general model for the confidence interval, sample under the null hypothesis for the test.

Part (b)(i). The probability formulae here are rather simple, if you can see them clearly, but in my experience students really struggle with this blue-sky probability modelling step, which is why the generous marks. I'm happy if students simply write down the answers. I'll give partial marks for a reasonable explanation that gives the wrong answers. For a placebo subject,

$$P(t, c) = \begin{cases} (1 - q)^t & \text{if } c = 0 \\ (1 - q)^{t-1}q & \text{if } c = 1 \end{cases}$$

This assumes that “develop the disease” is independent from day to day. Students have been told to assume random variables are independent unless otherwise specified, and to note down in their answers when they make the independence assumption.

Part (b)(i). For a vaccinated subject

$$P(t, c) = \begin{cases} f + (1 - f)(1 - q)^t & \text{if } c = 0 \\ (1 - f)(1 - q)^{t-1}q & \text{if } c = 1. \end{cases}$$

Part (b)(iii). The log likelihood is just

$$\log P(t_1, c_1, \dots, t_n, c_n) = \sum_i \log \text{appropriate } P \text{ term above.}$$

NOT ASKED FOR: how to do the estimation. This likelihood does not lend itself to an algebraic solution. We should run a numerical optimization, e.g. using `scipy`. Standard numerical optimization routines operate over unconstrained parameters, but here we need to force $q \in [0, 1]$ and $f \in [0, 1]$. The standard way to do this, as in lecture notes section 1.2, is to rewrite the model in terms of $q = e^{q'}1 + e^{q'}$ and $f = e^{f'}1 + e^{f'}$, and optimize over $q' \in \mathbb{R}$ and $f' \in \mathbb{R}$.
